

Mixed precision GPUs parallelism

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Recap

Highload

- Foundations
 - von Neumann architecture
 - Flynn's taxonomy
 - Data locality principle
- MapReduce paradigm
 - Hadoop stack
 - YTsaurus
- kNN indexes

Outline

GPUs, CUDA



Mixed precision

girafe
ai

01

Mixed precision

- <https://docs.nvidia.com/deeplearning/performance/mixed-precision-training/index.html>
- <https://arxiv.org/pdf/1710.03740>
- https://en.wikipedia.org/wiki/Half-precision_floating-point_format
- https://web.eecs.umich.edu/~justincj/slides/eecs498/WI2022/598_WI2022_lecture18.pdf
- <https://cloud.google.com/blog/products/ai-machine-learning/bfloat16-the-secret-to-high-performance-on-cloud-tpus>
- <https://www.youtube.com/watch?v=b5dAmcBKxHg>
- <https://www.youtube.com/watch?v=i1fIBtdhjlq>
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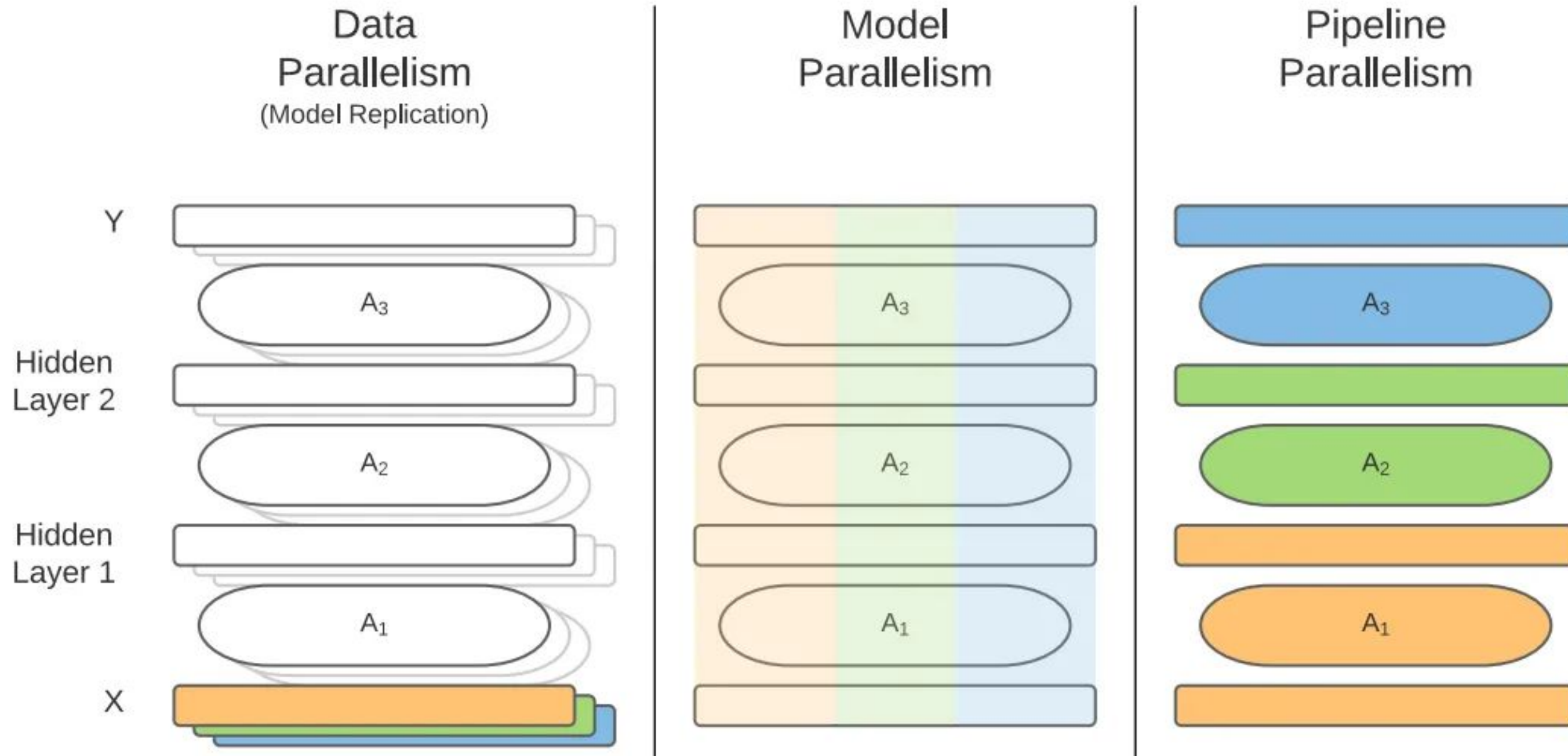
GPU parallelism

girafe
ai

02

Types of parallelization in ML

<https://betterprogramming.pub/parallel-and-distributed-training-in-deep-learning-for-beginners-part-1-introduction-612a4534a117>



Data parallelism

- https://pytorch.org/tutorials/intermediate/ddp_tutorial.html
- <https://pytorch.org/docs/stable/notes/ddp.html>
- https://pytorch.org/tutorials/beginner/ddp_series_multigpu.html
- DDP in lightning
 - https://github.com/girafe-ai/mllops/blob/25s_mipt/repo_example/cats-and-dogs/cats_and_dogs/train.py
- Fully Shared DP
 - https://huggingface.co/docs/accelerate/usage_guides/fsdp
 - https://docs.pytorch.org/tutorials/intermediate/FSDP_tutorial.html
- <https://www.youtube.com/watch?v=4i76hmmnJEo>

ZeRO

- <https://arxiv.org/pdf/1910.02054v3>
- <https://www.microsoft.com/en-us/research/blog/zero-deepspeed-new-system-optimizations-enable-training-models-with-over-100-billion-parameters/>
- <https://www.youtube.com/watch?v=tC01FRB0M7w>
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Fully Sharded Data Parallel

- https://docs.pytorch.org/tutorials/intermediate/FSDP_tutorial.html
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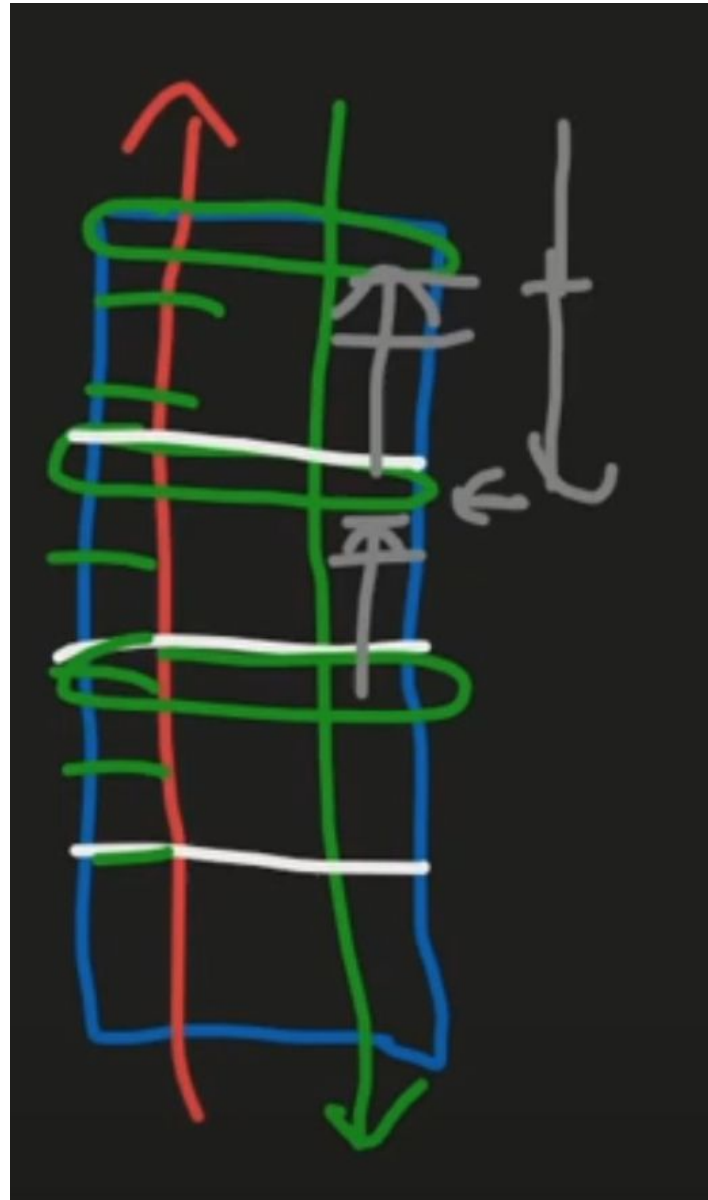
Pipeline parallelism

- <https://arxiv.org/pdf/1811.06965>
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Model parallelism

- <https://arxiv.org/pdf/1909.08053>
-

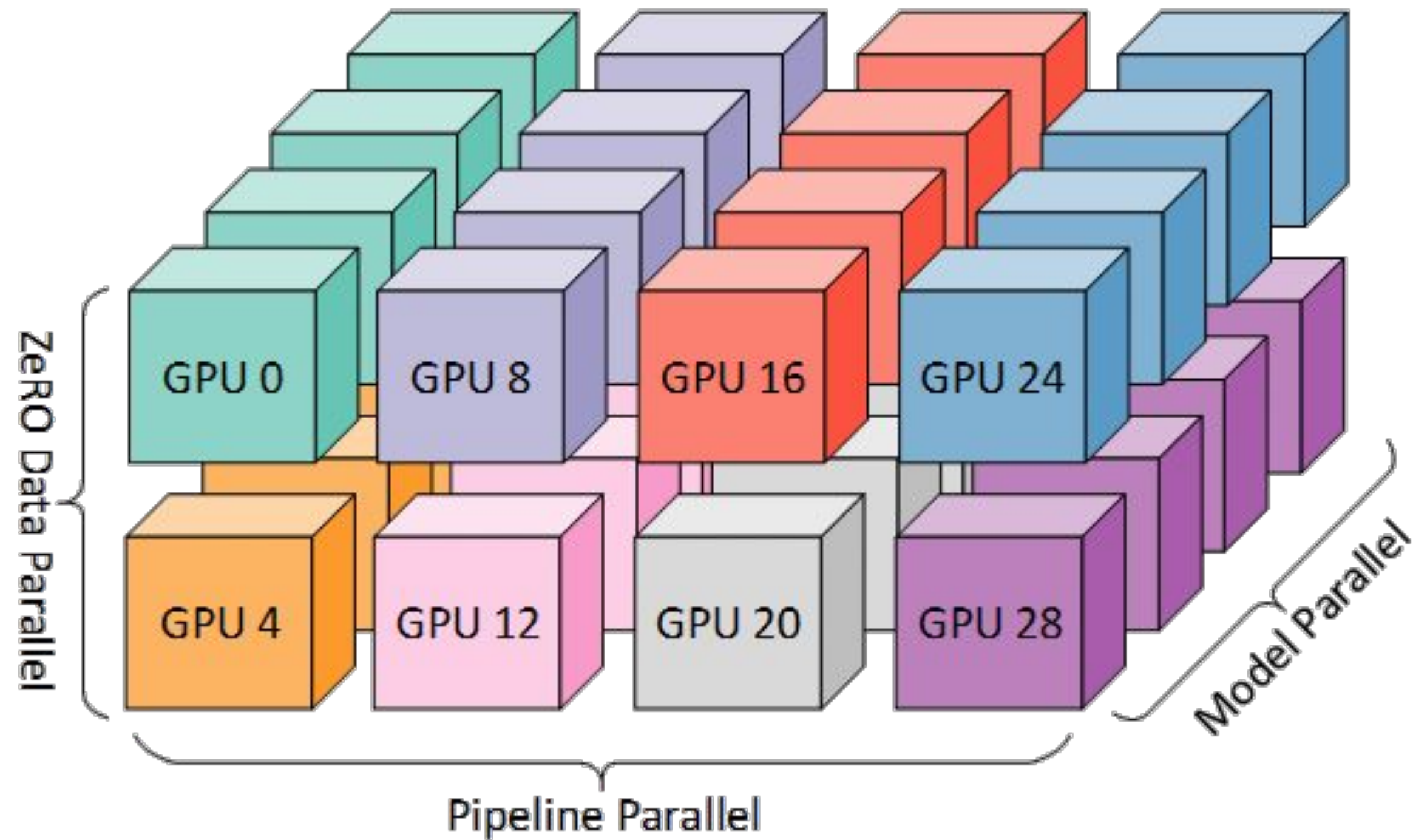
Activation checkpointing



Deep Speed

- <https://github.com/deepspeedai/DeepSpeed>
- <https://www.deepspeed.ai/>
-

3D GPU parallelism



More links

- Great videos on gpu parallelism
 - <https://www.youtube.com/watch?v=hc0u4avAkuM>
 - <https://www.youtube.com/watch?v=pTChDs5uD8I>
- <https://huggingface.co/docs/transformers/v4.15.0/en/parallelism>
- <https://www.microsoft.com/en-us/research/blog/deepspeed-extreme-scale-model-training-for-everyone/>
- <https://huggingface.co/spaces/nanotron/ultrascale-playbook>
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Revise

1. Problem statement
2. Metrics
3. Models

Thanks for attention!

Questions?

